

MOTOR PROJECT

Temperature Speed Controlled DC Motor

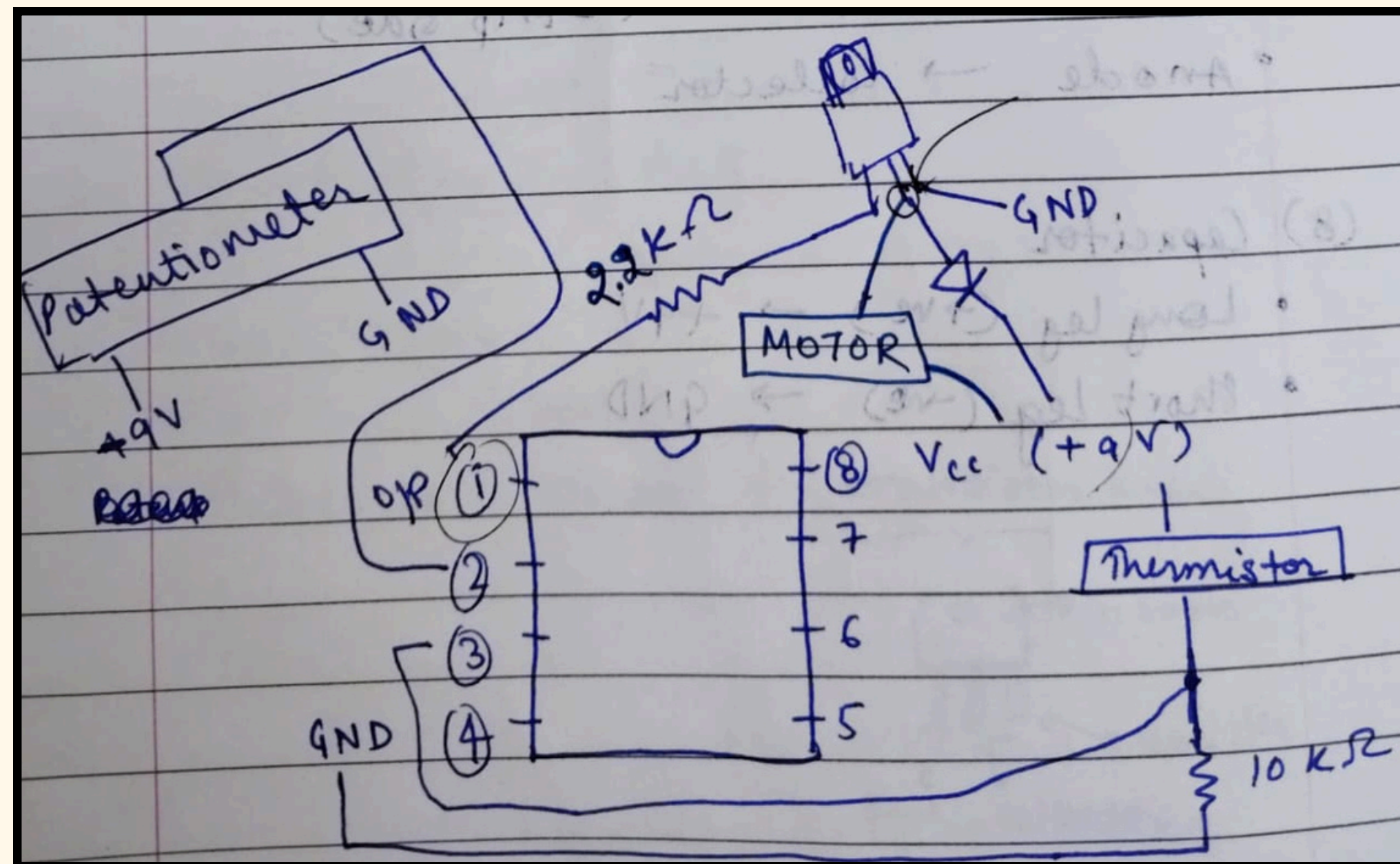
WORKING PRINCIPLE

- **Principle:** The system operates on the principle of **electromagnetic rotation**, i.e., a current-carrying conductor placed in a magnetic field experiences a force, producing rotation in a DC motor. The speed of the DC motor is controlled by varying the current through it, **based on temperature**.
- **Operation:**
 - **Temperature Sensing:** An **NTC (Negative Temperature Coefficient) thermistor** senses the surrounding temperature. As the temperature increases, its resistance decreases.
 - **Voltage Conversion:** This change in resistance is converted into a corresponding voltage variation using a **voltage divider network**.
 - **Signal Processing:** The **LM358 operational amplifier** processes this voltage by comparing it with a reference level set by a potentiometer.

WORKING PRINCIPLE

- **Current Amplification:** The resulting output signal drives a **TIP122 transistor**, which regulates the current supplied to the motor.
- Once the voltage exceeds the predefined threshold set by the Potentiometer, the transistor enters the "**Saturation**" state.
- **Motor Operation:** As the current flows, the **interaction** between the magnetic field and the current-carrying coil produces **torque**, causing the motor to rotate.
- **Speed Control:** As the temperature increases, the thermistor resistance decreases, the voltage increases, the current increases, the torque increases and hence the **motor speed increases**.

DESIGN OF MOTOR



VARIOUS PARTS AND MATERIALS

- Enamelled Copper Wire (5m) - ₹15/meter
- Neodymium Magnets (4) - ₹20 each
- Safety Pins (2) - Working as a commutator
- Insulated Copper Wire (1m) - ₹20/meter
- Breadboard - ₹50
- Thermistor (NTC 100k)
- Op-Amp (LM358)
- Transistor (TIP122)
- Diode (1N4007)
- 10k ohm resistor (1)
- 2.2k ohm resistor (1)
- Potentiometer (10k ohm)
- 100 uF capacitor (1)
- Heat sink (1)
- Electrical Tape and Double sided Tape
- Total cost - ₹440
- DC Supply - Available in Laboratory

PROBLEMS FACED AND SOLUTIONS

Coil not Rotating: No sufficient current was available through the battery and the resistance of the coil was very high. For the resistance part, we decreased the turns of the coil and for battery part, we used DC supply.

Coil sagging: Changed Shape to elliptical but did not work (flux leakage) so kept circular coil but reduced the number of turns and increased the thickness of the coil.

PROBLEMS FACED AND SOLUTIONS

Not properly magnetising: We have to keep the magnets close to the coil (2mm). This distance is between the lowest part of the coil and top most part of the magnets. Also the magnets had to be kept just below the centre of the coil.

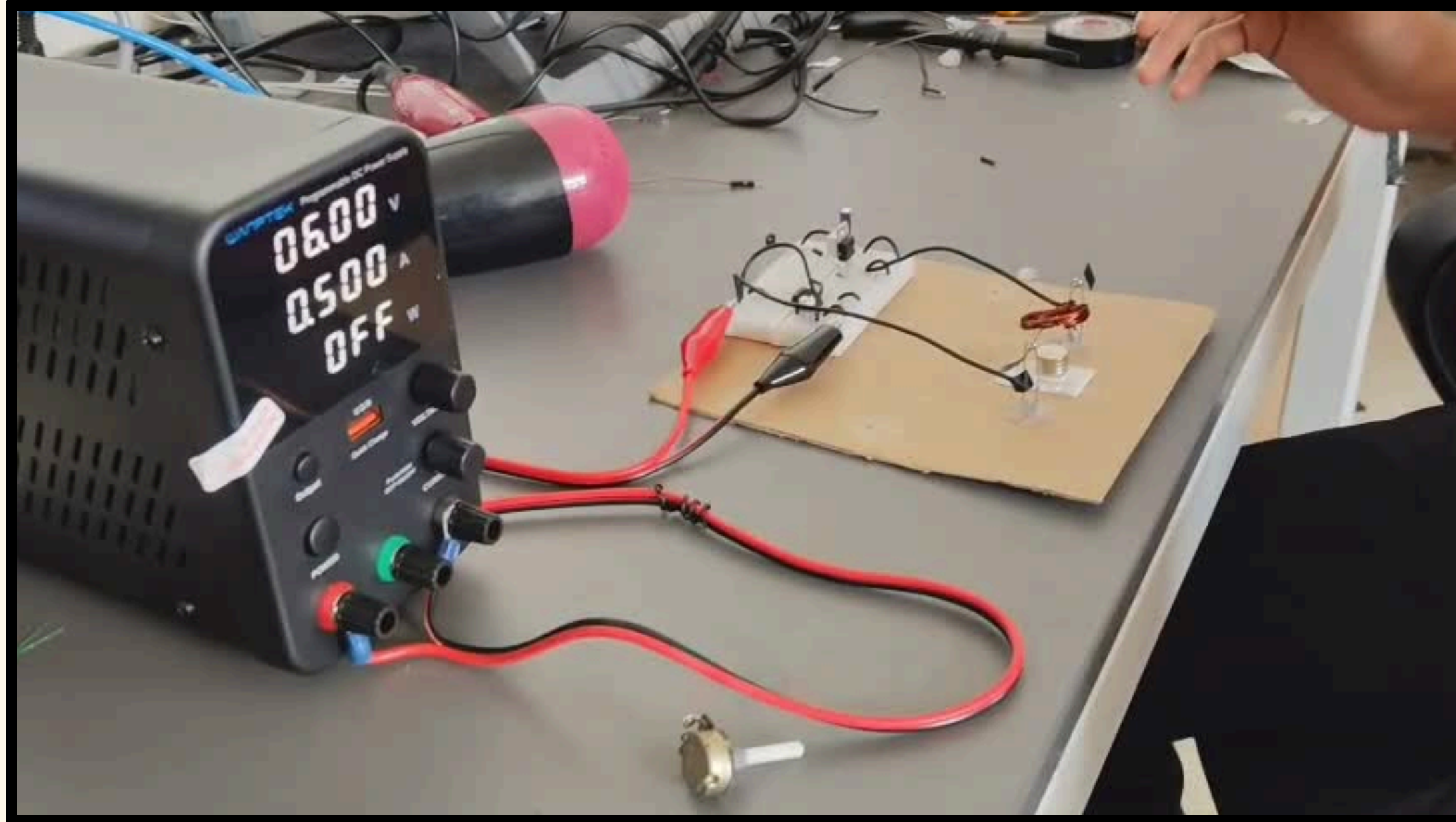
Coil rotating but Sparks created: Spark was due to contact and no contact of the coil with the commutator, it was controlled using a capacitor which made the motor stable and provided smoothening of the signals.

PROBLEMS FACED AND SOLUTIONS

Motor speed not under control: Initially the DC supply given was 9V, which made the motor start at a high speed due to which we were not able to judge the increase in speed due to temperature rise. So we decreased the supply to 6V.

Transistor Heating up: Reduced the base current by changing 1k ohm resistor with 2.2k ohm resistor. And attached a heat sink with the transistor so that it cools down faster.

DEMONSTRATION



**THANK
YOU**